SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding *(2)*	Method Selection *(2)*	Solution Process *(2.5)*	Accuracy of Calculation *(2)*	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I G SAI DEEPAK(192311432) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak G

Activity 4 : COT2 : Assement Tool - 1 Analytical problems

1. Suppose that the data for analysis includes that attribute age. The age values for the data tuples are (in creasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) What is the mean of the data and median?

(b) What is the mode of data? Comment on the datas modality.

(c) What is the midrange of data?

(d) Can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?

(e) Give the five number Summary of data.

(f) Show a boxplot of the data

(a)

A) (i) mean $= \frac{1}{n} \sum_{i=1}^{n} x_i$

$= \frac{809}{27} = 29.96 \approx 30$

(ii) Meadian (odd) $= \frac{27+1}{2} = 14^{th} \simeq 14 \Rightarrow 25$

(b) Mode: 25 & 35 are repeating Continouosly. as only 2 values are repeating. it is a bimodal.

(c) Midrange $= \frac{\text{max value} + \text{min value}}{2} = \frac{70+13}{2}$

$= \frac{83}{2} \Rightarrow 41.5$

(d) median = 14 So take first 13 values,

Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25.

median $= \frac{n+1}{2} = \frac{13+1}{2} = 7^{th}$ value $= 20$

$Q_1 = 20$

Q_3 = ignore median write other values till End.

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

median of $Q_3 = \frac{n+1}{2} = 7^{th}$ value

$$\boxed{Q_3 = 35}$$

(e) five number Summary: min, median, Q_1, Q_3, max.

min = 13, ~~median~~

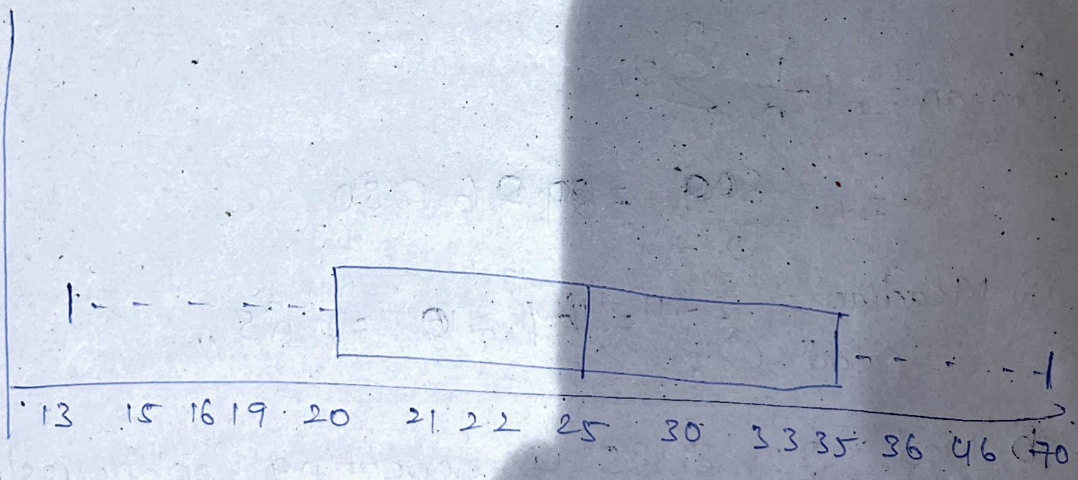
Q_1 = 20

median = 25

Q_3 = 35

max = 70

(f) Boxplot for the data



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(2) Given the following measurements for the variable age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28, Standardize the variable by the following:

Compute the mean absolute deviation of age!

Sol: Mean $= \frac{1}{n}\sum_{i=1}^{n} x_i \Rightarrow \frac{330}{10} = 33$

x_i (age)	$\lvert x_i - \bar{x} \rvert$
18	$\lvert 18-33 \rvert = 15$
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

Sum of absolute deviations $= 88$

Mean Absolute deviation (MAD)

$$MAD = \frac{\sum \lvert x - \bar{x} \rvert}{n}$$

$$= \frac{88}{10} = 8.8$$

③ Suppose the data for analysis include the attribute age. The age values are (in increasing order)

13, 15, 16, 16, 19, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Sol: Mean $= \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{809}{27} \simeq 30$

Bin depth : 3

Count on bin $= \frac{\text{occurrence}}{\text{bin depth}} = \frac{27}{3} = 9$

B_1 : 13, 15, 16 − 14.6

B_2 : 16, 19, 20 − 18.3

B_3 : 20 21, 22 − 21

B_4 : 22, 25, 25 − 24

B_5 : 25, 25, 30 − 26.6

B_6 : 33, 33, 35 − 33.6

B_7 : 35, 35, 35 − 35

B_8 : 36, 40, 45 − 40.3

B_9 : 46, 52, 70 − 56

Bin	Smooth values
B1	14.6, 14.6, 14.6
B2	18.3, 18.3, 18.3
B3	21, 21, 21
B4	24, 24, 24
B5	26.6, 26.6, 26.6
B6	33.6, 33.6, 33.6
B7	35, 35, 35
B8	40.3, 40.3, 40.3
B9	56, 56, 56

(4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000, (a) min-max normalization by setting min=0, and max = 1 (b) z- Score normalization.

Sol: (a)

min-max normalisation $(x') = \dfrac{x - \min(x)}{\max(x) - \min(x)}$

$x' = \dfrac{x - 200}{1000 - 200} \Rightarrow x' = \dfrac{x - 200}{800}$

Orginal	Calculation	Normalized value
200	200-200/800	0
300	300-200/800	0.125
400	400-200/800	0.25
600	600-200/800	0.5
1000	1000-200/800	1.0

$\boxed{0, 0.125, 0.25, 0.5, 1.0}$

(b) **Z-Score Normalization** : $Z = \frac{x-\mu}{\sigma}$

$$\mu = \frac{200+300+400+600+1000}{5} = \frac{2500}{5} = 500$$

Standard deviation :

x	$x-\mu$	$(x-\mu)^2$
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000

$\Sigma(x-\mu)^2 = 4,00,000$

$$\sigma^2 = \frac{\Sigma(x-\mu)^2}{5} = \frac{400000}{5} = 80,000$$

$$\sigma = \sqrt{80,000} = 282.84$$

Z-Score :

Original	Calculation	Z-Value
200	-300/282.84	-1.06
300	-200/282.84	-0.71
400	-100/282.84	-0.35
600	100/282.84	-0.35
1000	500/282.84	1.76

Z score : $\boxed{-1.06, -0.71, -0.35, -0.35, 1.76}$

5. Suppose a group of 12 Sales price records has been Sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into 3 bins by each of following methods. (a) Equal frequency partitioning (b) Equal width (c) Clustering.

Sol: $n = 12$ bins $= 3$

(a) Equal frequency partitioning:

Records per bin $= \frac{12}{3} = 4$ Records

Bin	Records
B_1	5, 10, 11, 13
B_2	15, 35, 50, 55
B_3	72, 92, 204, 215

(b) Equal width partitioning:

$$\text{width} = \frac{\text{max} - \text{min}}{\text{no. of bins}} = \frac{215 - 5}{3} = \frac{210}{3} = 70$$

The intervals:

Bin 1 : 5 → 74

Bin 2 : 75 → 144

Bin 3 : 145 → 215

Bin	Interval	Values
Bin 1	5 → 74 ~~5, 10, 11, 13, 15, 35~~	5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	75 → 144	92
Bin 3		

ⓒ Clustering partition :

Cluster (bin)	Values
Cluster 1	5, 10, 11, 13, 15
Cluster 2	35, 50, 55, 72, 92
Cluster 3	204, 215

⑥ Suppose the hospital tested the age and body fat for 18 randomly selected adults with the following result.

age ~~% fat~~

23
~~23~~
~~27~~
27
39
41
47
49
50
52
54
54
56
57
58
58
60
61

% fat

9.5
26.5
7.8
17.8
31.4
25.9
27.4
27.2
27.2
31.2
34.6
42.5
28.8
33.4
30.2
34.1
32.9
41.2
35.7

ⓐ Calculate mean, median & standard deviation of age & % fat

ⓑ draw boxplots for age & % fat

ⓒ Draw a scatter plot & a q-q plot based on these two variables.

ⓓ Normalize the two variables based on z-score normalization.

ⓔ Calculate the correlation coefficient (person's product moment coefficient). Are these two variables positively (or) negatively correlated?

Sol: (a) (i) mean $= \frac{1}{n} \sum_{i=1}^{n} x_i \Rightarrow \frac{836}{18}$ 46.44

(ii) Median $= \frac{n}{2}$ (even) $= \frac{18}{2} = 9$

Avg of 9 & 10^{th} term $= \frac{50+52}{2} = 51$

$S = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}} = 13.22$

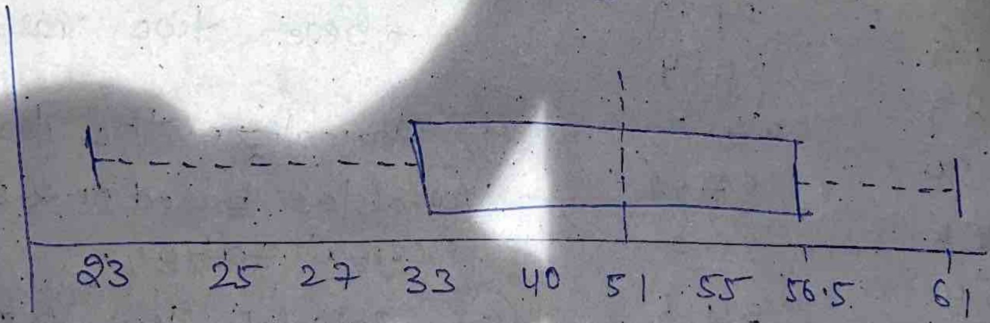
mean $= \frac{1}{n} \sum_{i=1}^{n} x_i = \frac{518.1}{18} = 28.76$

median $= \frac{30.2+31.2}{2} = 30.7$

$S = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}} = 9.25$

(b) (i) Boxplox : min = 23 ; Q_1 = 33, median = 51,

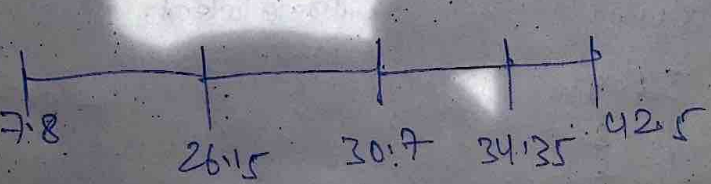
Q_3 = 56.5 , max = 61



(ii) % fat body values.

min = 7.8 ; Q_1 = 26.15 ; Median = 30.7 ; Q_3 = 34.35

max = 42.5



(c) Scatter plot : Plotted points.

(23, 9.5)	(54, 42.5)
(23, 26.5)	(54, 28.8)
(27, 7.8)	(56, 28.8)
(27, 17.8)	(56, 33.4)
(39, 31.4)	(57, 30.2)
(41, 25.9)	(58, 34.1)
(47, 27.4)	(58, 32.9)
(49, 27.2)	(60, 41.2)
(50, 31.2)	(61, 35.7)
(52, 34.6)	

(d)

Age	Z(Age)	% fat	Z% fat
23	-1.77	9.5	-2.08
23	-1.77	26.5	-0.25
27	-1.47	7.8	-2.27
27	-1.47	7.8	-2.27
39	-0.56	17.8	-1.19
41	-0.41	31.4	0.28
47	0.04	25.9	-0.31
49	0.19	27.4	-0.15
50	0.27	27.2	-0.17
52	0.42	31.2	0.26
54	0.57	34.6	0.63
54	0.57	42.5	1.48
54	0.72	28.8	0.00
56			

(e) Pearson's Correlation Coefficient :

$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2 \sum (y-\bar{y})^2}}$$

$$\boxed{r = 0.818}$$